The argument principle turns a sign condition on the imaginary axis into the uniform Penrose condition

Marnie Smith

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 1139-1146; its proof environment opens at line 1147, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 1147-1189 is the whole environment. The two labelled implications impliesPenrose and impliesPenrose0 used in the proof are declared inside the counted statement itself.

Editorial note. Four environments are retained as context: the uniform Penrose condition (source lines 203-205), which is the conclusion; the directional integral estimate gnorm (source lines 1037-1042); the integral representation Lingood-coordinates (source lines 1059-1062); and the boundedness and decay lemma (source lines 1092-1106).

Editorial note. This article carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label argumentprinciplelemma.

Editorial note. The retained preamble is the publisher layout adapter, the native font and layout package and the native mathematical macro and theorem block of the pinned source.

Editorial scope. One original result of the pinned publisher article, counted once: the argument-principle reduction of the uniform Penrose condition to a sign condition on the imaginary axis (author label argumentprinciplelemma).

Allowed standard prerequisites (editorial).

- Linearised operator symbol and its low-frequency limit. The symbol is an integral transform of the background, and its limit as the parameter tends to zero is an explicit integral transform evaluated at zero parameter.. They are the two objects on which the sign conditions are imposed.
- Uniform Penrose condition. The uniform Penrose condition requires that the real part of the symbol stay above a negative bound uniformly in the parameter, the spectral variable and the frequency direction.. It is the conclusion of the statement.

- Analyticity and the argument principle. The symbol is analytic in the half-plane of the spectral variable, and a curve lying in the image of the imaginary axis that avoids a half-line bounds the argument of the relevant factor.. It is the mechanism that converts the sign condition into the bound.

$$\inf_{\substack{\operatorname{Re}\lambda\geq 0\\ k\in\mathbb{R}^d}} \left|1+\widetilde{\mathscr{L}}(\lambda,k)\right| \geq c_{\mathrm{P}} \tag{1}$$

Lemma 0.1. *Let $n \geq 0$, and let α be a multi-index. Then, for every $k \neq 0$,*

$$\int_0^\infty \langle s\rangle^n \left|\partial_k^\alpha \widehat{g}\big(\tfrac{k}{|k|}s\big)\right| ds \lesssim_{d,\delta_0,n,|\alpha|} \|g\|_{H^{n+1/2+\delta_0}_{|\alpha|+\lceil d/2\rceil}}.$$

$$\widetilde{\mathscr{L}}(\lambda,k;\hbar) = \frac{2}{\hbar|k|}\widehat{w}(k)\int_0^\infty e^{-\lambda s/|k|}\sin(\tfrac{\hbar}{2}|k|s)\widehat{g}\big(\tfrac{k}{|k|}s\big)\,ds. \tag{2}$$

Lemma 0.2. *Suppose $w \in L^1(\mathbb{R}^d)$ and $g \in H^{3/2+\delta_0}_{\lceil d/2\rceil}(\mathbb{R}^d)$. Then:*

a. As $|k| \to \infty$, the quantity $\widetilde{\mathscr{L}}(\lambda,k;\hbar) \to 0$, with decay uniform in $\operatorname{Re}\lambda \geq 0$ and $\hbar \in [0,1]$.

b. There exists a constant $C_1 = C_1(d,\delta_0)$, independent of $\hbar \in [0,1]$, such that for all $\operatorname{Re}\lambda \geq 0$ and $k \in \mathbb{R}^d$,

$$\left|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\right| \leq C_1\|w\|_{L^1}\|g\|_{H^{3/2+\delta_0}_{\lceil d/2\rceil}}.$$

c. If in addition $g \in H^{3/2+\delta_0}_{1+\lceil d/2\rceil}(\mathbb{R}^d)$, then there exists a constant $C_2 = C_2(d,\delta_0)$, independent of $\hbar \in [0,1]$, such that for all $\operatorname{Re}\lambda \geq 0$ with $\lambda \neq 0$ and $k \in \mathbb{R}^d$,

$$\left|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\right| \leq C_2\frac{|k|}{|\lambda|}\|w\|_{L^1}\|g\|_{H^{3/2+\delta_0}_{1+\lceil d/2\rceil}}.$$

Lemma 0.3. *Suppose $w \in L^1(\mathbb{R}^d)$, $g \in H^{3/2+\delta_0}_{1+\lceil d/2\rceil}(\mathbb{R}^d)$, and that the following implications hold:*

$$\forall \hbar \in [0,1],\ \tau \in \mathbb{R},\ k \in \mathbb{R}^d, \quad \operatorname{Im}\widetilde{\mathscr{L}}(i\tau,k;\hbar) = 0 \implies \operatorname{Re}\widetilde{\mathscr{L}}(i\tau,k;\hbar) > -1, \tag{3}$$

and, for the low-frequency response,

$$\forall \tau \in \mathbb{R},\ \omega \in \mathbb{S}^{d-1}, \quad \operatorname{Im}\widetilde{\mathscr{L}}_0(i\tau,\omega) = 0 \implies \operatorname{Re}\widetilde{\mathscr{L}}_0(i\tau,\omega) > -1. \tag{4}$$

Then the uniform Penrose condition (1) is satisfied.

Proof. For each $\hbar \in [0,1]$ and $k \in \mathbb{R}^d$, the assumptions on w and g ensure that the map $\lambda \mapsto \widetilde{\mathscr{L}}(\lambda,k;\hbar)$ is analytic in the half-plane $\{\operatorname{Re}\lambda > 0\}$, continuous on $\{\operatorname{Re}\lambda \geq 0\}$ and, by Lemma 0.2 (c), tends to zero as $|\lambda| \to \infty$. By the argument

principle, the number of zeros of the function $\lambda \mapsto 1 + \widetilde{\mathscr{L}}(\lambda, k; \hbar)$ in $\{\operatorname{Re}\lambda > 0\}$ is equal to the winding number of the curve

$$\Gamma := \big\{\widetilde{\mathscr{L}}(i\tau, k; \hbar) : \tau \in \mathbb{R}\big\}$$

around the point -1. The implication (3) ensures that Γ does not meet the half-line $(-\infty, -1]$. It follows that $1 + \widetilde{\mathscr{L}}(\lambda, k; \hbar)$ has no zeros in $\{\operatorname{Re}\lambda \geq 0\}$. The same argument, applied to $\mu \mapsto 1 + \widetilde{\mathscr{L}_0}(\mu, \omega)$ for fixed $\omega \in \mathbb{S}^{d-1}$ and using (4), shows that $1 + \widetilde{\mathscr{L}_0}(\mu, \omega)$ has no zeros in $\{\operatorname{Re}\mu \geq 0\}$.

It remains to show that the infimum in (1) is positive. For $k \neq 0$, $\hbar \in [0, 1]$ and $\operatorname{Re}\lambda \geq 0$, the representation (2) reads

$$\begin{aligned}\widetilde{\mathscr{L}}(\lambda, k; \hbar) &= \widehat{w}(k)\, \Phi\left({}^{\lambda}\!/_{|k|}, \hbar|k|, {}^{k}\!/_{|k|}\right), \\ \Phi(\mu, h, \omega) &:= \int_0^\infty e^{-\mu s}\, \frac{2}{h} \sin\left({}^{hs}\!/_{2}\right) \widehat{g}(\omega s)\, ds,\end{aligned}$$

where the factor $2h^{-1}\sin(hs/2)$ is understood as s when $h = 0$, so that $\widetilde{\mathscr{L}_0}(\mu, \omega) = \widehat{w}(0)\Phi(\mu, 0, \omega)$. The function Φ is continuous on $\{\operatorname{Re}\mu \geq 0\} \times [0, \infty) \times \mathbb{S}^{d-1}$. Indeed, the integrand is continuous and bounded by $s\|\widehat{g}\|_{L^\infty}$, so the integral over $[0, R]$ is continuous for every $R > 0$, while $\left|2h^{-1}\sin(hs/2)\right| \leq s$ and the Cauchy–Schwarz and trace estimates in the proof of Lemma 0.1 give, uniformly in $\operatorname{Re}\mu \geq 0$, $h \geq 0$ and $\omega \in \mathbb{S}^{d-1}$,

$$\begin{aligned}\int_R^\infty \left| e^{-\mu s}\, \frac{2}{h} \sin\left({}^{hs}\!/_{2}\right) \widehat{g}(\omega s)\right| ds &\leq \left(\int_R^\infty \frac{s^2}{\langle s\rangle^{3+2\delta_0}}\, ds\right)^{1/2} \left(\int_0^\infty \langle s\rangle^{3+2\delta_0} |\widehat{g}(\omega s)|^2\, ds\right)^{1/2} \\ &\lesssim_{d,\delta_0} R^{-\delta_0} \|g\|_{H^{3/2+\delta_0}_{\lceil d/2\rceil}},\end{aligned}$$

so Φ is a uniform limit of continuous functions. Now set

$$F(\mu, r, \omega, \hbar) := 1 + \widehat{w}(r\omega)\, \Phi(\mu, \hbar r, \omega),$$

so that $F(\mu, r, \omega, \hbar) = 1 + \widetilde{\mathscr{L}}(r\mu, r\omega; \hbar)$ for $r > 0$ and $F(\mu, 0, \omega, \hbar) = 1 + \widetilde{\mathscr{L}_0}(\mu, \omega)$.

By Lemma 0.2, there exist constants $K > 0$ and $M > 0$ such that, for $k \neq 0$ and $\operatorname{Re}\lambda \geq 0$, if either (i) $|k| \geq K$, or (ii) $|\lambda| \geq M|k|$, then

$$\left|1 + \widetilde{\mathscr{L}}(\lambda, k; \hbar)\right| \geq \frac{1}{2}$$

uniformly in $\hbar \in [0, 1]$, while $\left|1 + \widetilde{\mathscr{L}}(\lambda, 0; \hbar)\right| = 1$. Writing $\lambda = r\mu$ and $k = r\omega$, the remaining region corresponds to the compact set

$$\mathcal{K} := \big\{(\mu, r, \omega, \hbar) :\, \operatorname{Re}\mu \geq 0,\, |\mu| \leq M,\, 0 \leq r \leq K,\, \omega \in \mathbb{S}^{d-1},\, \hbar \in [0, 1]\big\}.$$

On this compact set, the function F is continuous, since $\widehat{w}$ and Φ are, and it does not vanish, by the first paragraph. Hence

$$c_{\mathrm{P}} := \min\left\{{}^{1}\!/_{2},\, \min_{\mathcal{K}} |F|\right\} > 0$$

is a lower bound for $\left|1 + \widetilde{\mathscr{L}}(\lambda, k; \hbar)\right|$ over $\operatorname{Re}\lambda \geq 0$, $k \in \mathbb{R}^d$ and $\hbar \in [0, 1]$. This establishes the uniform Penrose condition (1). $\square$

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